

MERIDIAN TRENCH RESEARCH CONSORTIUM

ANNUAL OPERATIONS & RESEARCH REPORT · FISCAL YEAR 2025

HALCYON DEEP Station · Meridian Trench, South-Eastern Pacific Sector

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Executive Summary

Fiscal year 2025 marked the third full year of continuous occupation of HALCYON DEEP, the consortium's benthic research station moored on the axial terrace of the Meridian Trench at a depth of 5,120 metres. Across the reporting period the station supported a resident complement of up to twenty-two personnel, sustained 46 crewed submersible sorties, and catalogued 849 biological specimens, of which 61 are provisionally described as new to science.

Operational availability of the station reached 96.4 per cent, despite a partial flooding event in the Wet Porch during July (see Section 8) and a nine-day communications outage following damage to the surface relay buoy ALBATROSS-2. Total consortium expenditure for the year is summarised in [Section 6](#); the audited figure of €47.2 million reflects a 4.1 per cent underspend against the approved envelope.

Scientific highlights included the first time-series characterisation of the Grunwald Vent Field, the recovery of an intact mature colony of the ghost siphonophore *Apoemia spectra*, and the deployment of the second-generation CTD-9 sensor string. International participation broadened materially, with new instrument contributions from partners in Japan, Germany, France, Russia and India, described in Section 7.

The Governing Board has approved a two-year extension of station operations through fiscal 2027, conditional on completion of the tether-replacement programme and ratification of the revised ANDON PROTOCOL for emergency crew ascent.

— *Dr. Ingrid Halvorsen, Consortium
Director*

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1 • Station Overview

HALCYON DEEP is a five-module pressurised habitat assembled between 2021 and 2022 and permanently stationed on the axial terrace of the Meridian Trench. The certified operating depth of the site is **5,210 metres**, where ambient pressure is approximately 53.4 MPa and water temperature holds near 1.9 °C year-round. Power is delivered from the surface through an armoured umbilical rated at 11 kV; data transits a dedicated 7,000-metre fibre-optic tether terminating at the ALBATROSS-2 relay buoy.

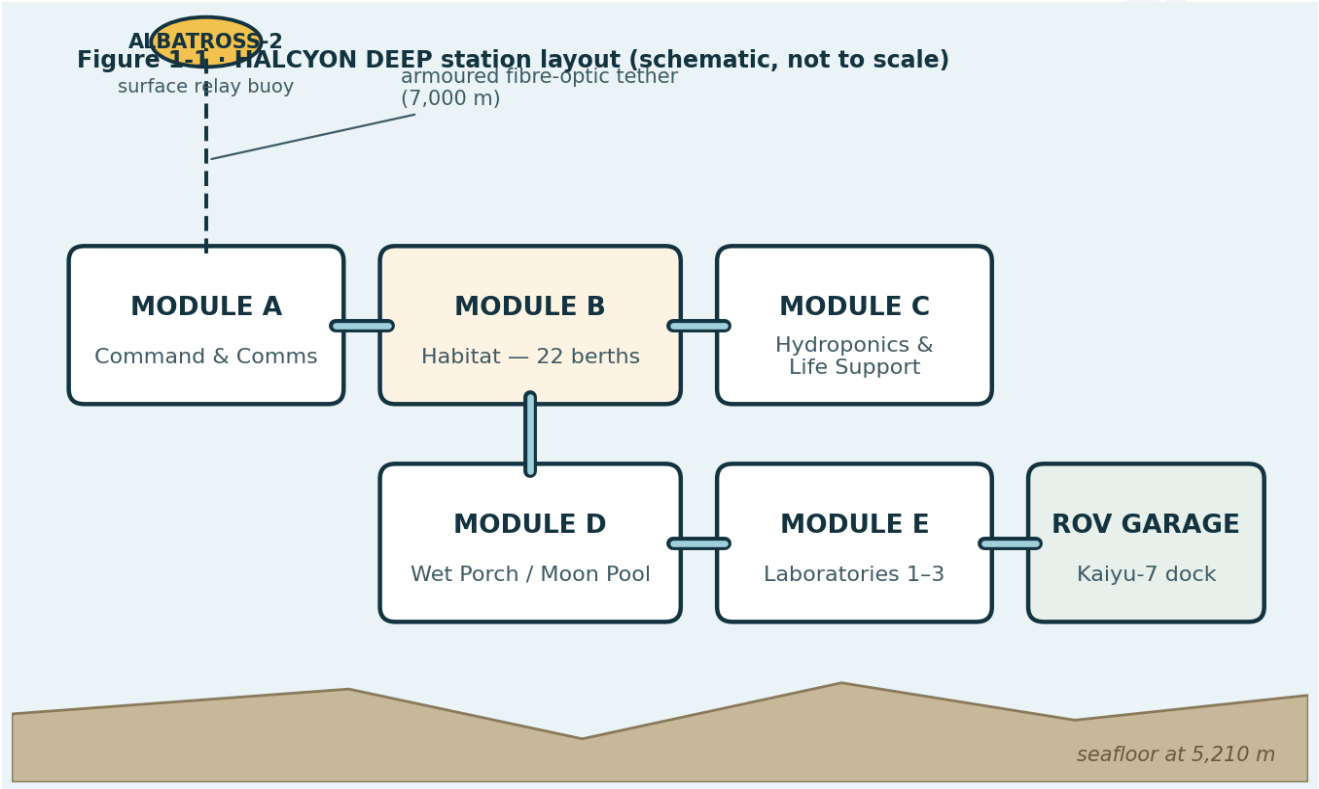


Figure 1-1. Station layout. Module functions and berth capacity are annotated on the schematic.

1.1 Structural certification

Parameter	Value	Certifying body	Valid to
Hull design pressure	62 MPa	Bureau Abyssal International (BAI)	2031-06
Operating depth (certified)	5,210 m	BAI	2031-06
Moon-pool cycle rating	12,000 openings	BAI	2029-01
Life-support endurance (isolated)	21 days	Consortium internal	annual

Table 1-1. Certification summary. The 62 MPa hull rating corresponds to the pressure housings described in Section 7.2.

2 • Personnel & Organisation

Resident staffing followed a two-crew rotation of nominally eleven persons each, exchanged at twelve-week intervals via the surface tender. Leadership appointments for the year are shown in Figure 2-1; the departmental roster follows in Table 2-1.

Figure 2-1 · Consortium leadership, FY2025

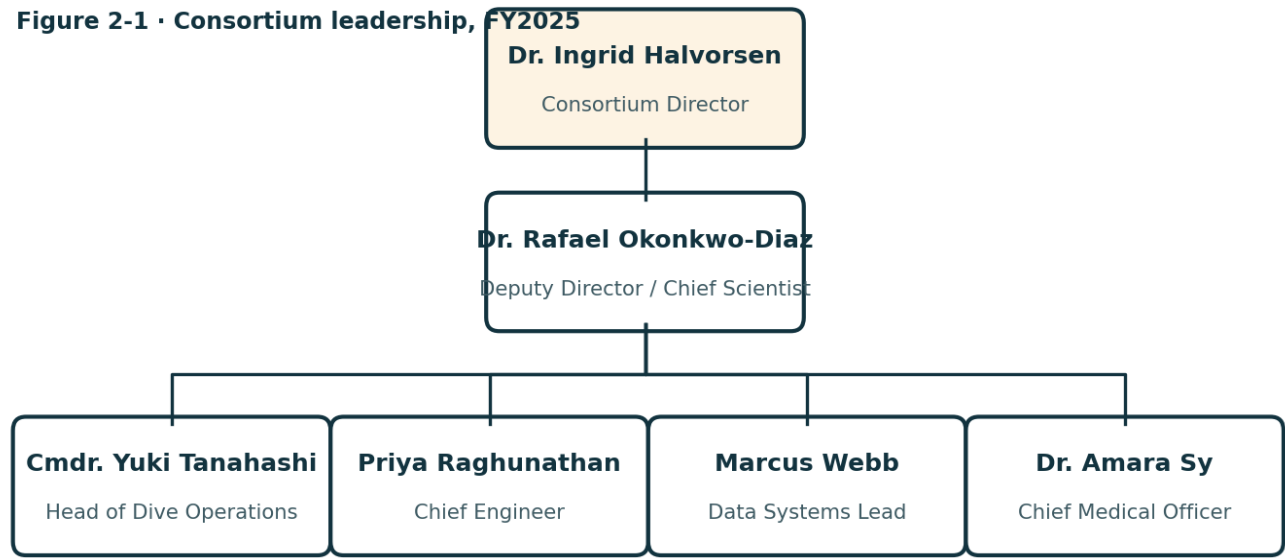


Figure 2-1. Consortium leadership. Names and roles appear only in this chart.

Department	Role	FTE	Rotation
Dive Operations	Submersible pilots	4	A / B
	Tether & winch technicians	3	A / B
	ROV supervisor ¹	1	A only
Science	Benthic ecologists	3	A / B
	Geochemists	2	B only
Engineering	Life-support engineers	3	A / B
	Data systems	2	A / B
Medical	Physician / hyperbaric	1	A / B

Table 2-1. Departmental roster (merged cells indicate shared department).

¹ Post seconded from the Kaiyu-7 operating partner for the duration of FY2025; see Section 7.1.

3 · Dive Operations

The submersible *Nereid-2* completed all crewed sorties for the year. Utilisation was highest in the third quarter, when **fourteen crewed dives** supported the Grunwald Vent Field time-series; the full log appears in Table 3-1. Dives D-2025-031 through D-2025-036 constitute the vent-field survey block referenced throughout Section 4.

3.1 First quarter

Eleven sorties concentrated on tether inspection and the photogrammetry mosaic of the southern terrace. Weather aloft was benign; no dive was aborted. Average bottom time was 4.6 hours, constrained by battery conditioning cycles carried over from the winter layup.

3.2 Second quarter

Twelve sorties supported sediment coring and the first larval-trap rotation. The nine-day ALBATROSS-2 outage (incident IR-2025-03) forced two dives to operate on acoustic-only telemetry, a contingency rehearsed but never previously used in live operations.

3.3 Third quarter

The vent-field time-series dominated the quarter. Back-to-back sorties on 12–14 August achieved the first 72-hour continuous instrument watch at the Grunwald site. The manipulator stall on D-2025-034 (incident IR-2025-06) cost one station-keeping day.

3.4 Fourth quarter

Seven sorties closed out the year, biased toward recovery and winterisation tasks. The final dive of the season, D-2025-046, recovered all three larval traps and the acoustic beacon string ahead of the January maintenance stand-down.

Dive ID	Date (2025)	Pilot	Max depth (m)	Duration (h:mm)	Objective	Sam ples	Δ plan (min)
D-2025-001	03 Jan	K. Osei	5,150	4:50	Specimen collection	11	+4
D-2025-002	05 Feb	L. Fernández	4,380	7:05	Tether inspection	2	(6)
D-2025-003	07 Mar	M. Duval	4,110	7:20	Photogrammetry mosaic		+4
D-2025-004	09 Jan	Y. Tanahashi	5,204	6:05	Sensor string service	0	+4
D-2025-005	11 Feb	K. Osei	5,204	3:05	Sediment coring	0	+15
D-2025-006	13 Mar	L. Fernández	4,110	7:50	Larval trap recovery		+9
D-2025-007	15 Jan	M. Duval	4,110	7:20	Acoustic beacon swap	2	(12)
D-2025-008	17 Feb	Y. Tanahashi	4,620	7:05	Vent field time-series	6	0
D-2025-009	19 Mar	K. Osei	4,620	3:20	Specimen collection	2	+4
D-2025-010	21 Jan	L. Fernández	4,380	7:05	Tether inspection	6	+9
D-2025-011	23 Feb	M. Duval	5,208	7:50	Photogrammetry mosaic	—	0
D-2025-012	03 Apr	Y. Tanahashi	5,208	7:50	Sensor string service	2	0
D-2025-013	05 May	K. Osei	4,890	4:20	Sediment coring		(6)
D-2025-014	07 Jun	L. Fernández	5,050	7:50	Larval trap recovery	2	+15
D-2025-015	09 Apr	M. Duval	5,208	5:05	Acoustic beacon swap		(6)
D-2025-016	11 May	Y. Tanahashi	5,204	4:35	Vent field time-series	0	(12)
D-2025-017	13 Jun	K. Osei	5,204	3:05	Specimen collection	—	(6)
D-2025-018	15 Apr	L. Fernández	5,204	6:42	Tether inspection	9	(6)
D-2025-019	17 May	M. Duval	5,150	5:35	Photogrammetry mosaic	6	(12)
D-2025-020	19 Jun	Y. Tanahashi	5,208	3:05	Sensor string service	2	(12)
D-2025-021	21 Apr	K. Osei	4,380	3:35	Sediment coring	11	(6)
D-2025-022	23 May	L. Fernández	5,208	5:50	Larval trap recovery	11	0
D-2025-023	25 Jun	M. Duval	4,110	6:35	Acoustic beacon swap	0	(6)
D-2025-024	03 Jul	Y. Tanahashi	4,380	6:05	Vent field time-series	0	
D-2025-025	05 Aug	K. Osei	5,050	4:20	Specimen collection	4	(12)
D-2025-026	07 Sep	L. Fernández	5,208	3:20	Tether inspection	4	(12)
D-2025-027	09 Jul	M. Duval	5,050	4:50	Photogrammetry mosaic	—	(6)
D-2025-028	11 Aug	Y. Tanahashi	5,050	6:35	Sensor string service	11	(12)
D-2025-029	13 Sep	K. Osei	4,890	4:05	Sediment coring	0	+9
D-2025-030	15 Jul	L. Fernández	4,890	4:05	Larval trap recovery	4	
D-2025-031	17 Aug	M. Duval	4,620	5:35	Vent field time-series		+9
D-2025-032	19 Sep	Y. Tanahashi	5,204	7:35	Vent field time-series	6	(6)
D-2025-033	21 Jul	K. Osei	5,150	4:05	Vent field time-series	4	
D-2025-034	23 Aug	L. Fernández	5,204	6:50	Vent field time-series	4	+4
D-2025-035	25 Sep	M. Duval	5,208	6:05	Vent field time-series	0	+4
D-2025-036	27 Jul	Y. Tanahashi	4,890	6:20	Vent field time-series		0
D-2025-037	04 Aug	K. Osei	4,110	3:05	Sediment coring	6	+9
D-2025-038	06 Sep	L. Fernández	4,380	5:05	Larval trap recovery		

Dive ID	Date (2025)	Pilot	Max depth (m)	Duration (h:mm)	Objective	Sam ples	Δ plan (min)
D-2025-039	08 Jul	M. Duval	4,890	7:50	Acoustic beacon swap	0	+15
D-2025-040	03 Oct	Y. Tanahashi	5,050	5:35	Vent field time-series	4	+4
D-2025-041	05 Nov	K. Osei	4,380	6:50	Specimen collection	4	(12)
D-2025-042	07 Dec	L. Fernández	5,050	3:20	Tether inspection		+15
D-2025-043	09 Oct	M. Duval	5,150	5:50	Photogrammetry mosaic	—	+15
D-2025-044	11 Nov	Y. Tanahashi	4,620	7:05	Sensor string service	0	(6)
D-2025-045	13 Dec	K. Osei	5,150	4:05	Sediment coring	—	(6)
D-2025-046	15 Oct	L. Fernández	5,050	3:35	Larval trap recovery	6	0

Table 3-1. Complete FY2025 dive log (46 sorties). Negative schedule variances shown in parentheses; blank cells indicate data not recorded.

Following dive D-2025-042 the recovery winch reported intermittent load-cell dropouts, and as a precaution the tether was rewound at reduced speed to avoid

birdcaging of the armoured strands — a failure mode in which individual wires buckle outward under sudden tension release. Inspection at the surface confirmed no permanent deformation, and the winch controller firmware was subsequently patched (Section 5.2).

4 · Scientific Findings

4.1 Specimen catalogue

The FY2025 catalogue closed at 849 accessioned specimens across six phyla; the distribution is charted in Figure 4-2. Sixty-one specimens carry provisional new-species designations pending ratification.³

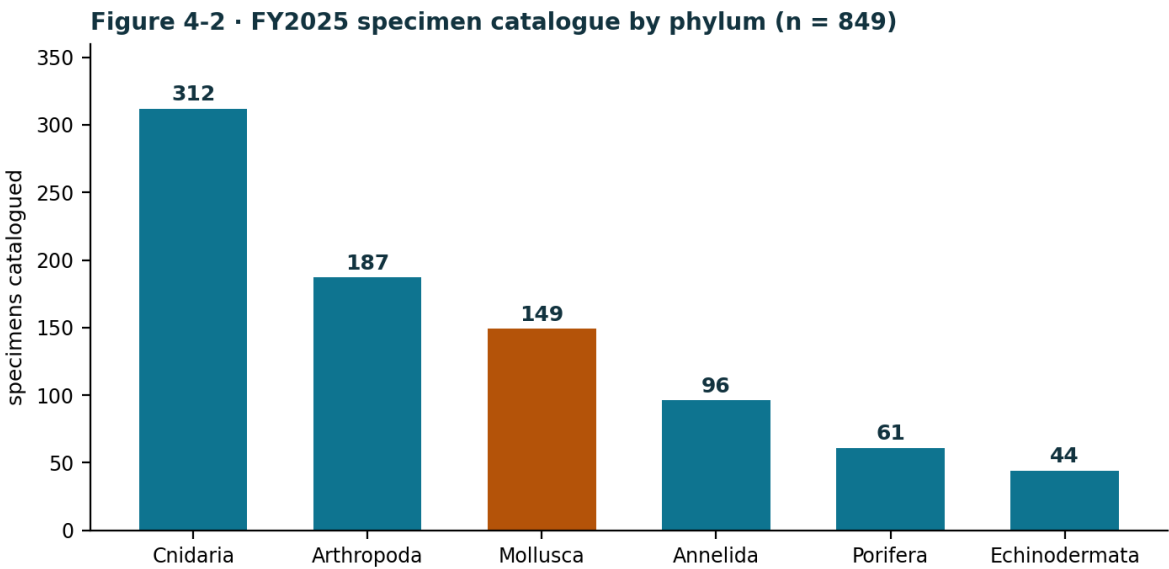
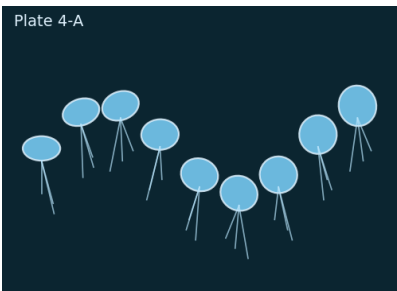


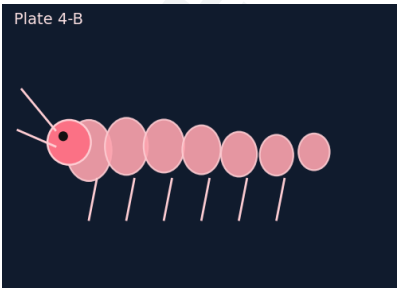
Figure 4-2. Catalogue distribution by phylum. Counts are given only in the chart.

³ New-species files were lodged with the Registry of Hadal Taxa under submission #RHT-2025-0917 on 4 December 2025.

4.2 The ghost siphonophore



Juvenile colonies of *Apolemia spectra*, the ghost siphonophore, were observed on eleven occasions along the trench wall between 4,600 and 5,100 metres. Juvenile colonies average **4.2 metres** in length and drift with siphosome contracted, feeding on amphipod swarms attracted to the vent halo.



Mature colonies of *Apolemia spectra*, the ghost siphonophore, were observed on three occasions along the trench wall between 4,600 and 5,100 metres. Mature colonies reach **6.8 metres** in length and drift with siphosome extended, feeding on amphipod swarms attracted to the vent halo. Plate 4-B shows the commensal amphipod *Hirondellea grunwaldi* recovered from a mature colony.

Plates 4-A / 4-B. Field sketches (stylised). Note the deliberately parallel phrasing of the two paragraphs above: only life stage, sighting count, length and posture differ.

4.3 Water column & dissolved oxygen

The CTD profile from cast MT-25-118 (Figure 4-3) confirms the near-isothermal deep layer below 3,500 metres. Dissolved-oxygen saturation was computed with the consortium's standard fit, whose coefficients are tabulated in Appendix B (Table B-1):

$$\ln C_{O_2}^{sat} = A_1 + A_2 \left(\frac{100}{T} \right) + A_3 \ln \left(\frac{T}{100} \right) + S \left[B_1 + B_2 \left(\frac{T}{100} \right) \right]$$

where T is absolute temperature (K), S is practical salinity, and $C_{O_2}^{sat}$ is expressed in mLL^{-1} . Hydrostatic pressure follows $P = P_0 + \rho gh$; at station depth this yields 53.4 MPa using the constants of Table B-1.

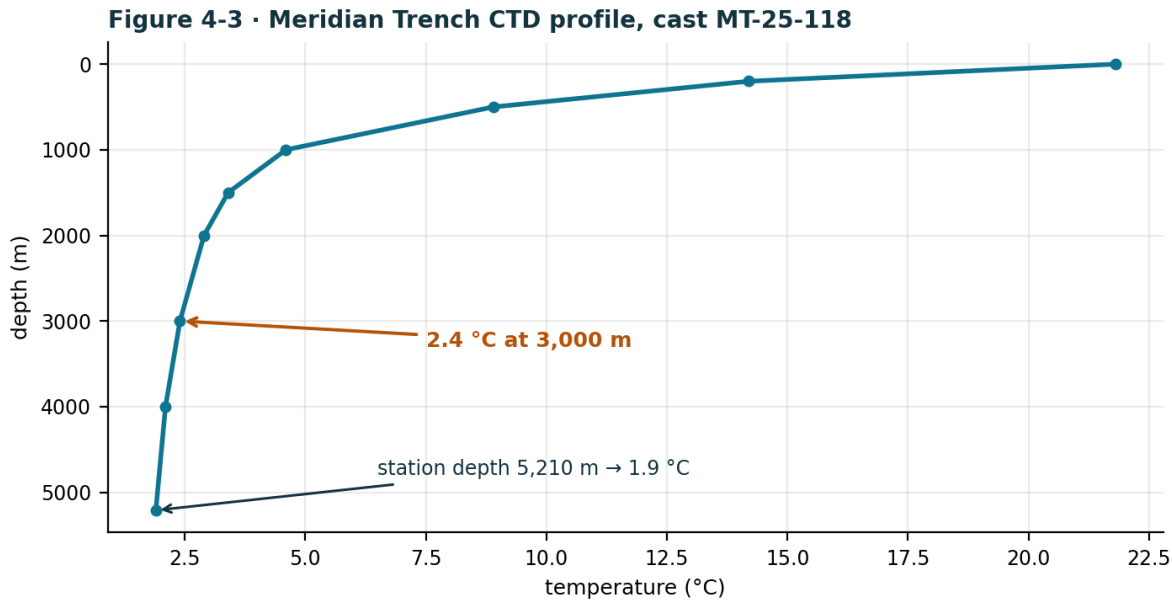


Figure 4-3. CTD temperature profile. Spot values are annotated only on the chart.

4.4 Grunwald Vent Field time-series

Monthly maxima of exit-fluid temperature at the principal orifice (marker GVF-A) are tabulated below. The July gap reflects the Wet Porch flooding stand-down; the October spike followed a minor seismic swarm recorded by the consortium hydrophone array.

Month	Max exit-fluid temp (°C)	Plume rise (m)	H ₂ S (mmol/kg)	Watch dives
January	287	41	5.9	1
February	291	44	6.1	2
March	288	42	6.0	1
April	290	43	6.2	2
May	286	40	5.8	2
June	289	41	6.0	1
July	—	—	—	0
August	294	47	6.6	6
September	292	45	6.4	4
October	301	52	7.1	2
November	293	46	6.3	1
December	290	44	6.1	1

Table 4-1. GVF-A monthly maxima. The em-dash row marks the July observation gap.

5 · Sensor & Software Systems

5.1 CTD-9 acquisition service

The acquisition daemon samples the CTD-9 string and raises a hard alarm above the certified envelope. Operational constants are set in firmware, not configuration:

```
MODULE = "ctd9d" # telemetry acquisition daemon v3.4.1
SAMPLE_RATE_HZ = 250 # per channel
DEPTH_ALARM_M = 5300 # hard ceiling – DO NOT EDIT
CHANNELS = ["cond", "temp", "pres", "o2", "turb"]

def frame(sample: dict) -> bytes:
    payload = pack("<5f", *[sample[c] for c in CHANNELS])
    return HDR + payload + crc16(payload)
```

5.2 Winch controller CRC

The winch firmware (patched after the D-2025-042 anomaly) frames telemetry with CRC-16/CCITT:

```
/* winchctl v2.7 – crc16.c (excerpt) */
uint16_t crc16(const uint8_t *p, size_t n) {
    uint16_t crc = 0xFFFF;
    while (n--) {
        crc ^= (uint16_t)(*p++) << 8;
        for (int i = 0; i < 8; i++)
            crc = (crc & 0x8000) ? (crc << 1) ^ 0x1021 : (crc << 1);
    }
    return crc;
}
```

5.3 Pre-dive software checklist

- Verify acquisition daemon heartbeat
 - ctd9d --status returns OK on all five channels
 - clock skew to surface < 40 ms
- Winch controller
 - load-cell self-test (three cycles)
 - CRC error counter zeroed
- ANDON PROTOCOL panel — lamp test, both crews

6 · Budget & Finance

Expenditure is reported in thousands of euros. The consolidated position for the year appears in Table 6-1; category shares are charted in Figure 6-1. Variances in parentheses are unfavourable.

Category	FY2024 Actual	FY2025 (€ thousands)		
		Budget	Actual	Variance
Vessel Operations	15,210	16,400	15,940	460
Personnel	11,870	12,100	12,310	(210)
Research Programmes	8,020	9,000	8,660	340
Equipment & Maintenance	4,760	5,600	5,120	480
Data & Communications	2,050	2,300	2,240	60
Contingency	980	2,400	1,530	870
Total	42,890	47,800	45,800	2,000

Table 6-1. Consolidated expenditure. Multi-level header: the three right-hand columns are grouped under FY2025.

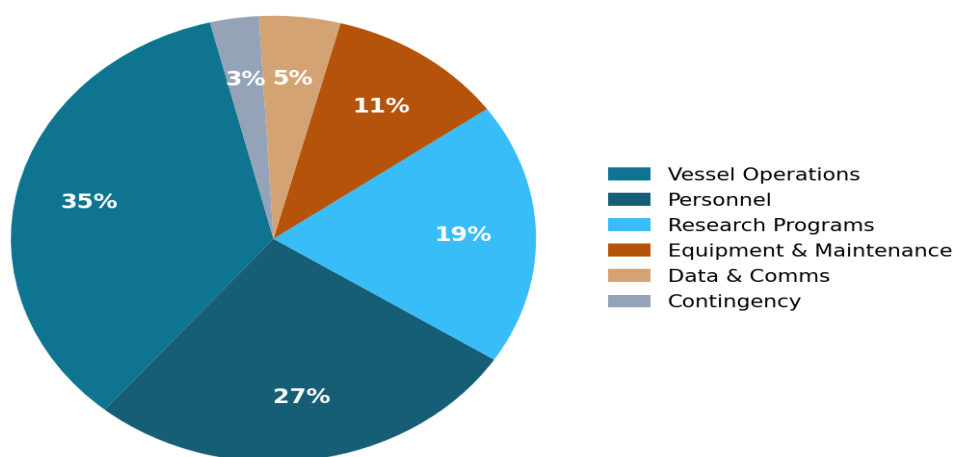
Figure 6-1 · FY2025 expenditure shares

Figure 6-1. FY2025 expenditure shares. Percentages appear only in the chart.

7 · International Partnerships

Partner statements are reproduced in the language of submission, as required by the consortium charter. English summaries are deliberately omitted.

7.1 Japan — 深海技術機構 (Shinkai Gijutsu Kiko)

深海技術機構は本年度、遠隔操作無人探査機「海遊七号」をコンソーシアムに提供した。本機の定格潜航深度は六五〇〇メートル（6,500 m）であり、ハルシオン・ディープ基地のROVガレージに常駐配備されている。運用要員一名が二〇二五年度を通じて基地に出向した。

7.2 Deutschland — Polarforschungsinstitut Bremerhaven

Das Polarforschungsinstitut Bremerhaven lieferte im Berichtsjahr sämtliche Titan-Druckgehäuse der zweiten Generation. Jedes Gehäuse ist für einen Betriebsdruck von **62 MPa** ausgelegt und wurde einzeln im Druckkessel des Instituts abgenommen. Die Übergabe erfolgte am 11. März 2025 in Bremerhaven.

7.3 France — Institut Océanographique de Roscoff-Sud

L'Institut Océanographique de Roscoff-Sud a fourni **trois réseaux de sonar** à ouverture synthétique du modèle « Écho-Marin 9 », installés sur le flanc nord de la terrasse axiale. La portée nominale de chaque réseau est de 1800 mètres.

7.4 Россия — Институт глубоководных исследований

Институт предоставил судно снабжения ледового класса Arc5 «Северная Звезда», выполнившее четыре рейса в течение отчётного года. Судно обеспечивает доставку тяжёлого оборудования в зимний период.

7.5 India — Chennai logistics partner (भारत)

मानसून अवधि (जून-सितम्बर) के दौरान सतही पुनःआपूर्ति हर छह सप्ताह में केवल एक बार संभव होती है। इस कारण स्टेशन पर खाद्य एवं ईंधन का भंडार न्यूनतम बारह सप्ताह के स्तर पर रखा जाता है। आपूर्ति पोत का संचालन चेन्नई स्थित साझेदार संस्था द्वारा किया जाता है।

Statement 7-5 is reproduced as a facsimile image of the submitted text (Devanagari script).

8 · Incident Log (fold-out)

ID	Date	Location	Classification	Description	Root cause	Downtime	Status
IR-2025-01	09 Feb	Module C	Minor	Hydroponics pump cavitation during filter swap	Air ingress at quick-disconnect	3 h	Closed
IR-2025-02	27 Mar	Tether	Moderate	Fibre attenuation spike, comms degraded to 40 Mbps	Micro-bend at splice joint S-11	31 h	Closed
IR-2025-03	02 Jun	ALBATROSS-2	Major	Surface buoy struck by drifting long-line gear; nine-day comms outage	Third-party fishing gear	216 h	Closed
IR-2025-04	17 Jul	Module D — Wet Porch	Major	Partial flooding of Wet Porch during moon-pool cycling; compartment sealed and pumped	Failed penetrator seal, P/N HX-1147	62 h	Closed
IR-2025-05	17 Jul	Module D	Minor	Secondary bilge sensor false-positive during IR-2025-04 recovery	Condensation on connector	—	Closed
IR-2025-06	30 Aug	Nereid-2	Moderate	Manipulator wrist stalled at depth on dive D-2025-034	Hydraulic contamination	48 h	Closed
IR-2025-07	11 Oct	Winch	Minor	Load-cell dropouts on recovery of D-2025-042	Firmware race condition (patched v2.7)	6 h	Closed
IR-2025-08	05 Dec	Module A	Minor	UPS transfer glitch during generator changeover	Aged capacitor bank	1 h	Open — parts on order

Table 8-1. FY2025 incident register. This page is intentionally landscape-oriented.

The founding survey memorandum of 1998 is reproduced overleaf as an archival scan; note that its preliminary depth estimate predates modern multibeam correction.

MEMORANDUM - EXPEDITION FILE 98/117

DATE: 14 March 1998
FROM: Dr. Elias Grunwald, Principal Investigator
VESSEL: RV Thalassa Verne (survey leg 3)
SUBJECT: Preliminary sounding, unnamed trench axis

Single-beam echosounder returns across the axial valley give a best preliminary depth estimate of 5,340 metres (uncorrected for sound-velocity profile). Bottom character: soft pelagic ooze over indurated substrate. Two dredge hauls recovered manganese-crusting cobbles and one intact hexactinellid skeleton.

Recommend the feature be entered in the survey ledger under the working name MERIDIAN TRENCH pending hydrographic ratification.

Weather forced early termination of leg 3 after cyclone warning 09-B. All hands well.

(signed) E. Grunwald

9 · Equipment Requisition (specimen form)

A completed requisition is reproduced below in its native form layout.

Requisition No.	REQ-7741
Date raised	03 November 2025
Requested by	M. Webb — Data Systems Lead
Item description	Armoured fibre-optic tether, 7,000 m spool, type FOT-7K/AR
Quantity	2
Unit cost	€18,400
Total	€36,800
Priority	☒ Urgent ☐ Routine ☐ Deferred
Budget line	Equipment & Maintenance (Table 6-1)
Approved by	Dr. I. Halvorsen — Consortium Director <i>signature on file</i>

Form MTRC-F-12 Rev 4. Checked boxes are rendered as ballot glyphs; unchecked boxes remain empty.

10 · Data Management & Archive

All telemetry, imagery and acoustic data transit the fibre tether to the surface archive in near-real time and are mirrored to the consortium repository within 24 hours. Retention periods are governed by the schedule in Table 10-1; the ingest pipeline configuration is reproduced verbatim below.

```
{
  "pipeline": "mtrc-ingest",
  "version": "2.11.0",
  "chunk_size_kb": 512,
  "compression": "zstd:9",
  "checksum": "blake3",
  "mirrors": ["surface-a", "surface-b", "shore-archive"],
  "embargo_days": { "imagery": 180, "specimen_genetics": 365 }
}
```

Data class	Raw retention	Derived products	Access tier
CTD & sensor telemetry	Permanent	Permanent	Open after QC
Raw acoustic (hydrophone)	25 years	Permanent	Member
Video & still imagery	10 years	Permanent	Embargo 180 d
Specimen genetic sequences	Permanent	Permanent	Embargo 365 d
Engineering & housekeeping	7 years	5 years	Internal

Table 10-1. Retention schedule adopted by Governing Board resolution GB-2024-19.

11 · Environmental & Regulatory Compliance

Station operations proceed under three standing instruments issued by the (fictional) International Hadal Stewardship Commission. All were in good standing at year end. Benthic disturbance within the survey polygon remained at 38 per cent of the permitted annual footprint, and no discharge exceedances were recorded.

Instrument	Reference	Issued	Expires
Benthic Disturbance Permit	BD-2025-114	01 Jul 2025	30 Jun 2027
Specimen Collection Licence	SC-2023-388	15 Mar 2023	14 Mar 2028
Acoustic Emissions Consent	AE-2024-072	09 Sep 2024	08 Sep 2026

Table 11-1. Standing regulatory instruments, FY2025.

An application to extend the survey polygon 2.2 kilometres north-east, encompassing the secondary vent cluster identified in October, was lodged on 18 December 2025 and remains under review. The Commission has requested a supplementary acoustic-impact model, which the data systems team expects to deliver in the second quarter of fiscal 2026.

SPECIMEN COPY — INTERNAL

Appendix A · Glossary

ANDON PROTOCOL — Consortium emergency procedure authorising immediate crew ascent without surface concurrence when two independent hull sensors alarm within 90 seconds.

Axial terrace — The flat bench on the trench wall on which HALCYON DEEP is moored.

Birdcaging — Failure mode of wire rope in which strands buckle outward after sudden tension release.

CTD — Conductivity-Temperature-Depth sensor package; the station operates the CTD-9 string.

Grunwald Vent Field — Hydrothermal vent cluster 1.4 km north of the station, named for the 1998 survey PI.

Halocline — Depth interval of rapid salinity change; weakly expressed at the station site.

Moon pool — Open wet interface in Module D through which the submersible is launched.

Penetrator — Pressure-tight electrical or optical feed-through in a hull; see incident IR-2025-04.

RHT — Registry of Hadal Taxa — the (fictional) body ratifying new-species designations.

Siphosome — The feeding/reproductive chain of a siphonophore colony.

Tender — The surface support vessel performing crew rotation and resupply.

Umbilical — Armoured cable delivering 11 kV power from the surface installation.

Vent halo — Zone of chemosynthetic productivity surrounding an active vent.

Wet Porch — Colloquial name for the Module D moon-pool compartment.

Appendix B · Site Constants & Calibration

Values below are the consortium reference set for FY2025 and are cited from Sections 1 and 4.

Quantity	Symbol	Value	Notes
Station coordinates	—	12°44.8′ S, 78°03.2′ W	WGS-84
Certified station depth	h	5,210 m	multibeam-corrected, 2023
Local gravity	g	9.803 ms ⁻²	site survey
Mean seawater density (in-situ)	ρ	1.046 × 10 ³ kgm ⁻³	at station depth
Surface pressure	P ₀	101.3 kPa	standard
CTD-9 conductivity cell constant	κ	0.9821 ± 0.0004	calibration 2025-01-19
O ₂ fit coefficient	A ₁	−173.4292	Section 4.3 equation
	A ₂	249.6339	≈ dimensionless

Table B-1. Reference constants. The merged first-column cell groups the two oxygen-fit coefficients.

Appendix C · Specimen Accession Register (abridged)

Twenty-eight representative accessions from the FY2025 register. The full register (849 entries) is held in the shore archive. Holotypes are flagged in the final column.

Accession	Taxon (provisional)	Collected	Dive	Locality	Storage	Holo type
MTRC-B-0400	<i>Apolemia spectra</i>	18 Nov	D-2025-03 3	GVF-A halo	tank 2	
MTRC-B-0403	<i>Hirondellea grunwaldi</i>	27 Jun	D-2025-04 2	GVF-B flank	jar 08	
MTRC-B-0406	<i>Osedax sp. nov. 3</i>	21 Nov	D-2025-03 7	axial terrace		
MTRC-B-0409	<i>Bathyceramaster sp.</i>	11 Apr	D-2025-01 6	trench wall 4,800 m	tank 2	
MTRC-B-0412	<i>Cladorhiza abyssalis</i>	27 Apr	D-2025-01 4	sediment plain	cryo C-4	
MTRC-B-0415	<i>Munidopsis sp.</i>	22 Oct	D-2025-00 7	moon-pool bycatch	tank 2	●
MTRC-B-0418	<i>Peniagone sp.</i>	02 May	D-2025-04 4	GVF-A halo	cryo C-4	
MTRC-B-0421	<i>Careproctus sp. nov. 1</i>	03 Apr	D-2025-03 8	GVF-B flank	jar 03	
MTRC-B-0424	<i>Apolemia spectra</i>	02 Jun	D-2025-01 7	axial terrace	cryo C-4	
MTRC-B-0427	<i>Hirondellea grunwaldi</i>	20 Sep	D-2025-03 4	trench wall 4,800 m	tank 2	
MTRC-B-0430	<i>Osedax sp. nov. 3</i>	11 Jun	D-2025-01 7	sediment plain		
MTRC-B-0433	<i>Bathyceramaster sp.</i>	04 Feb	D-2025-03 6	moon-pool bycatch		
MTRC-B-0436	<i>Cladorhiza abyssalis</i>	19 Aug	D-2025-03 4	GVF-A halo	tank 2	
MTRC-B-0439	<i>Munidopsis sp.</i>	26 Aug	D-2025-01 0	GVF-B flank	jar 11	
MTRC-B-0442	<i>Hirondellea grunwaldi</i>	02 Aug	D-2025-01 9	axial terrace	jar 17	●
MTRC-B-0445	<i>Careproctus sp. nov. 1</i>	05 Apr	D-2025-00 9	trench wall 4,800 m	tank 2	
MTRC-B-0448	<i>Apolemia spectra</i>	02 Oct	D-2025-02 3	sediment plain	jar 03	
MTRC-B-0451	<i>Hirondellea grunwaldi</i>	03 Jun	D-2025-00 5	moon-pool bycatch	jar 08	
MTRC-B-0454	<i>Osedax sp. nov. 3</i>	14 Oct	D-2025-03 5	GVF-A halo		
MTRC-B-0457	<i>Bathyceramaster sp.</i>	22 Apr	D-2025-03 1	GVF-B flank	cryo C-4	
MTRC-B-0460	<i>Cladorhiza abyssalis</i>	04 Aug	D-2025-01 7	axial terrace	jar 11	
MTRC-B-0463	<i>Munidopsis sp.</i>	15 Sep	D-2025-02 4	trench wall 4,800 m	jar 03	
MTRC-B-0466	<i>Peniagone sp.</i>	24 May	D-2025-01 2	sediment plain	jar 08	
MTRC-B-0469	<i>Careproctus sp. nov. 1</i>	16 Feb	D-2025-01 1	moon-pool bycatch	jar 03	

Accession	Taxon (provisional)	Collected	Dive	Locality	Storage	Holo type
MTRC-B-0472	<i>Apolemia spectra</i>	19 May	D-2025-03 6	GVF-A halo		
MTRC-B-0475	<i>Hirondellea grunwaldi</i>	08 Nov	D-2025-01 7	GVF-B flank	cryo C-4	
MTRC-B-0478	<i>Osedax sp. nov. 3</i>	14 Oct	D-2025-01 3	axial terrace		
MTRC-B-0481	<i>Bathyceramaster sp.</i>	08 Oct	D-2025-01 2	trench wall 4,800 m	tank 2	

Table C-1. Abridged accession register. A filled circle marks holotype material; blank storage cells indicate material consumed by analysis.

Appendix D · Document Revision History

Rev	Date	Author	Change summary
A	2026-01-12	Secretariat	Initial draft for Board review
B	2026-02-02	R. Okonkwo-Diaz	Science sections revised; Figure 4-3 replaced
C	2026-02-22	I. Halvorsen	Approved for controlled distribution

Table D-1. Revision history for MTRC-AR-2025.

Colophon. This document is a wholly fictional synthetic artefact generated to stress-test retrieval-augmented generation pipelines. It contains five documented factual contradictions, data that exists only inside raster images, one simulated archival scan, one landscape fold-out, five languages, and a sentence split across a page boundary. The companion answer key enumerates all of them. Any resemblance to real institutions, vessels or persons is coincidental.